

Class #10 - Linear Independence

1. For any matrix $A \in \mathbf{R}^{m \times n}$, the equation $Ax = 0$ is _____.

Fill in the blank: $Ax = 0$ has a unique solution if and only if none of the columns is a _____ of the remaining columns.

2. Def: Given $u_1, \dots, u_p$ in $\mathbf{R}^n$, we say that *the set* $\{u_1, \dots, u_p\}$ _____ *linearly independent* if the equation

$$x_1 u_1 + \dots + x_p u_p = 0$$

has only the _____ solution.

If the equation has a nontrivial solution, then we say that the set $\{u_1, \dots, u_p\}$ _____ *linearly* _____.

3. Determine whether the following set is linearly independent or dependent.

(a) $\{0\}$

(b) $\left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \right\}$

Any set containing a single nonzero vector is _____.

(c) Any set containing the zero vector is _____.

(d) The set $\{u_1, u_2\}$ where u_1 and u_2 are independent is _____, otherwise, the set is _____.

(e) A set of vectors where more vectors are given than there are entries in each vector is _____.

(f) A set of vectors where fewer vectors are given than there are entries in each vector is _____.

(g) The set $\{v_1, \dots, v_4\}$ where $v_i \in \mathbf{R}^4$, and $v_3 \notin \text{Span}\{v_1, v_2, v_4\}$ is _____.

(h) The set $\{v_1, \dots, v_4\}$ where $\{v_1, v_2, v_3\}$ is linearly independent is _____.

4. Recall: Any matrix transformation $T(x) = Ax$ is a _____ transformation.

Question: Given an arbitrary linear transformation $T : \mathbf{R}^n \rightarrow \mathbf{R}^m$, can we find a matrix A so that $T(x) = Ax$?

5. Example: $T : \mathbf{R}^2 \rightarrow \mathbf{R}^2$ that rotates points (about the origin) through $\pi/3$ radians (counter-clockwise) is a linear transformation. Why?

Note: Any point in $\mathbf{R}^2$ can be written as a linear combination of _____ and _____.

So, $T(x_1, x_2) = ______ = [T(e_1) \quad T(e_2)]x$.

$T(1, 0) =$

$T(0, 1) =$

Hence, this rotation mapping can be written as Ax where $A = ______$.

6. Theorem: Let $T : \mathbf{R}^n \rightarrow \mathbf{R}^m$ be a linear transformation. Denote the columns of _____ matrix by $e_i, i = 1, \dots, ______$.

Then, $T(x) =$

The _____ matrix $[T(e_1) \ T(e_2) \ \cdots \ T(e_n)]$ is called the *standard matrix of T* .

7. Fill in the blanks:

(a) A linear transformation $T : \mathbf{R}^m \rightarrow \mathbf{R}^n$ is onto (that is, $\text{Range } T = \text{_____}$) if and only if its standard matrix has a pivot position in every _____.

(b) A linear transformation $T : \mathbf{R}^m \rightarrow \mathbf{R}^n$ is one-to-one (that is, if $u \neq v$ in Domain, then _____ in Range)
if and only if
 $T(x) = 0$ has _____ solution,

i.e., its standard matrix has a pivot position in every _____,

i.e., the set of columns of the standard matrix $\{T(e_1), T(e_2), \cdots, T(e_n)\}$ is _____.

- (c) A linear transformation $T : \mathbf{R}^n \rightarrow$ _____ is one-to-one and onto if and only if its standard matrix has a pivot position in every _____ and _____.